STA304H1 Term Test 1
June 4, 2026 2:30 PM to 4:30 PM

Print your information in the following table:

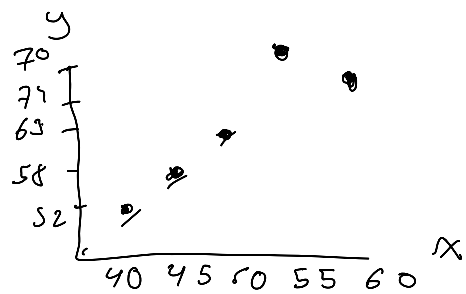
- Your STA304H1 term test 2 will be graded out of 33 points. This consists of 4 long answer questions (Q1 6 marks, Q2 8 marks, Q3 3 marks, Q4 6 marks) and 10 multiple choice questions (1 mark each).
- Answer the following questions in the space provided on pages with a QR code. Any work that is not on a page with a QR code will not be graded.
- Please write clearly using a pencil with a HB or darker lead. Do NOT use erasable pens. Regular pens with a dark ink are acceptable.
- To earn the full credit, you must show all steps/work that demonstrate the logical steps and course concepts you are applying in your problem solving. Final answers are worth at most 1 point.
- You are NOT allowed to refer to any resources (including an electronic calculator).
- You must complete the test individually and are NOT allowed to discuss with others during the test.
- If the question asks for a numeric answer (*these are starred*), your final answer should be written as a single expression involving only numbers and standard arithmetic operations. That is, a calculator should be sufficient to compute the final value. For example, you may leave your answer as $2e^3 \times \sqrt{8 \times 2.1} \times \log(4/3)$.

Part A: Long Answer (23 marks)

Show all work for full credit.

Q1. (6 marks) A simple random sample of $n = 5$ farms is selected from a population of $N = 100$ farms in Ontario. Let y be the crop yield (tons) and x be the farm land size (area). The sample data are given here:

Farm	x_i	y_i
1	40	52
2	55	70
3	60	74
4	45	58
5	50	63
Total	250	317



The population mean acreage is known to be 48.

a) Justify whether the proportionality assumption for ratio estimation is reasonable in this context. [1 mark]

This assumption is reasonable since the relationship between x and y appears linear and could reasonably go through the origin since crop yield $= 0$ when farm land size $= 0$ [+0.5 mark linear, +0.5 origin]

b) *Estimate Ontario's mean crop yield using ratio estimation.* [3 mark]

$$\hat{\bar{y}}_r = \hat{B}\bar{x}_U = \frac{(52+70+74+58+63)}{(40+55+60+45+50)} \times 48$$

[1 mark correct formula; 1 mark $\hat{B}$; 1 mark pop mean]

c) *Estimate the total crop yield in Ontario using ratio estimation.* [2 mark]

$$\hat{t}_{yr} = \hat{B}t_x = \hat{B} \times \bar{x}_U \times N$$

$$= \hat{B} \times 48 \times 100$$

[1 mark $t_x = 48 \times 100$; 1 mark formula]

Q2. **(8 marks)** An SRS of 1,500 licensed boat owners in a state was sampled from a list of 400,000 names with currently licensed boats. In the sample, 30% of licensed boat owners own a large motorboat (domain 1) and 70% of sampled licensed boat owners do not own a large motorboat (domain 2). Suppose that in the population, 20% of licensed boat owners own a large motorboat and 80% don't own a large motorboat. The following frequency table gives the number of respondents in each domain who reported having that number of children.

Number of Children	Domain 1	Domain 2
0	76	552
1	136	189
2	156	225
3	63	76
4	19	8
Total	450	1050

a) *Estimate the percentage of boat owners who have children in each domain.* [2 marks]

$$\hat{p}_1 = \frac{450-76}{450} \qquad \hat{p}_2 = \frac{1050-552}{1050}$$

1 mark each

b) *Estimate the average number of children per household among boat owners in each domain* [2 marks]

$$\hat{y}_1 = \frac{0\times76 + 1\times136 + 2\times156 + 3\times63 + 4\times19}{450}$$

$$\hat{y}_2 = \frac{0\times552 + 1\times189 + 2\times225 + 3\times76 + 4\times8}{1050}$$

1 mark each

c) *Estimate the mean number of children per household among all licensed boat owners using post-stratification.*

[2 marks]

$$\bar{y}_{post} = 0.2 \times \bar{y}_1 + 0.8\,\bar{y}_2$$

1 mark for correct weights

1 mark for correct formula

d) Explain why post-stratification may improve the accuracy of the estimate in part (c) compared to a simple average of all observations. Refer to the specific context of this problem, and numbers if necessary, in your answer.

[2 marks]

1mark discrepancy in proportions

the sample proportion of large boat owners (30%) differs from the population proportion (20%)

$$\bar{y}_s = 0.3\,\bar{y}_1 + 0.7\,\bar{y}_2$$

1 mark for comparison to a simple sample average

without poststratification the sample mean over-represents the averge # children among large boat owners when estimation the population mean children among all boat owners

Q3. (3 marks) In a one-stage cluster sampling design, show that $\hat{t}$ is an unbiased estimator of the population total. Clearly justify each step in your derivation and be sure to define any notation you introduce. A hint is available to you in MCQ7.

$$\hat{t} = N\bar{t}_S = \frac{N}{n}\sum_{i \in S} t_i$$

$$= \frac{N}{n}\sum_{i=1}^{N} I_i t_i \quad ; \text{ where } I_i = \begin{cases} 1 & \text{psu } i \in S \\ 0 & \text{otherwise} \end{cases}$$

Note some students might then write $t_i = \sum_{j=1}^{M} y_{ij}$ and that's fine as well.

$$E[\hat{t}] = \frac{N}{n}\sum_{i=1}^{N} E[I_i]\, t_i$$

$$= \frac{N}{n}\sum_{i=1}^{N} \frac{n}{N}\, t_i$$

$$= \sum_{i=1}^{N} t_i$$

1 mark: Understanding of random variables in the context of sampling theory. (t_i and y_{ij} are fixed but I_i is a random variable)

1 mark: Introduction of a random variable with goal of changing $\sum_{i \in S}$ to $\sum_{i=1}^{N}$

1 mark: $\sum_{i=1}^{N} t_i = t$ or $\sum_{i=1}^{N}\sum_{j=1}^{M_i} y_{ij} = t$

Q4. (6 marks) A market research firm took an SRS of 4 cities from the 45 cities in a region, and then took an SRS of m_i supermarkets from the M_i supermarkets in sampled city i. The following table gives the number of cases of apples sold during May in each of the sampled supermarkets.

City	M_i	m_i	Number of Cases Sold	Sample average of Cases Sold
1	52	10	146, 180, 251, 152, 72, 181, 171, 361, 73, 186	162.8
2	19	4	99, 101, 52, 121	93.25
3	8	2	12, 23	17.50
4	39	8	226,129,57,46,86,43,85,164	104.50
Total	118	24		

a) *Estimate the total number of cases sold in the region. Denote your estimate as $\hat{t}$. State what estimator you are using for this purpose.*

[3 marks]

total estimator from 2-stage design

(0.5 mark)

$$\hat{t} = \sum_{i \in S} \sum_{j \in S_i} \frac{N}{n} \times \frac{M_i}{m_i} y_{ij}$$ (0.5 marks)

$$= \frac{45}{4} \sum_{i \in S} M_i \times \frac{1}{m_i} \sum_{j \in S_i} y_{ij}$$

$$= \frac{45}{4} \sum_{i \in S} M_i \bar{y}_i$$

$$= \frac{45}{4} \left[52 \times 162.8 + 19 \times 93.25 + 8 \times 17.50 + 39 \times 104.50 \right]$$

* Some students will go through this a bit differently.

2 marks awarded for calculation steps

→ TA to decide Rubric

b) *Estimate the total number of supermarkets in the region.* [2 marks]

$$\hat{M}_0 = N\bar{M}_s$$

1 mark formula

$$= 45\,(52 + 19 + 8 + 39)/4$$

1 mark correct numbers

c) *Estimate the mean number of cases sold per supermarket in the region. You can use $\hat{t}$ in your answer instead of rewriting the estimate you obtained in a).* [1 marks]

$$\hat{\bar{y}} = \frac{\hat{t}}{\hat{M}_0}$$

← ok if they leave like this

$\boxed{\text{1 mark}}$

Part B: Multiple Choice (12 marks)

To answer the following multiple choice questions, fill in the corresponding rectangle on the provided Crowdmark bubble sheet. **Make sure the MCQ number matches the question number on the bubble sheet.**

MCQ1. A school randomly selects 10 classrooms and surveys every student in those classrooms. What type of sampling is this?

(A) Two-stage cluster sampling

(B) Systematic random sampling

(C) Stratified random sampling

(D) One-stage cluster sampling

(E) Simple random sampling

MCQ2. A researcher is conducting a survey to estimate the population mean income. They know that earned income is highly correlated with geographic status. They know from recent census data that exactly 22% of the target population resides in rural areas, 48% in suburban areas, and 30% in urban areas. However, their available sampling frame completely lacks any indicators for geographic status. Which of the following approaches (sampling design or analysis) is most appropriate for this scenario?

(A) A simple random sample from the frame, record each respondent's geographic status during data collection, and use the census proportions to weight the resulting post-strata means.

(B) Use optimal allocation to deliberately oversample the urban zone.

(C) Implement a domain estimation framework, which treats the sample sizes within each geographic zone as fixed constants across all theoretical sample iterations.

(D) Apply a standard combined ratio estimator using sample data to calculate auxiliary quantity t_x.

(E) Take a stratified random sample with rural, suburban, and urban as the strata to guarantee a proportional allocation.

MCQ3. In survey sampling, what is the fundamental structural difference between a stratum and a domain?

(A) Domains are used exclusively in cluster sampling designs, while strata are used in simple random sampling.

(B) Stratum sizes are always equal across the populations, whereas domains can vary in size.

(C) Strata are defined before sampling and used in the sample design, whereas domains are defined for analysis and their sample sizes are typically random.

(D) A domain requires knowing the population size of every subpopulation before sampling while a stratum does not.

(E) Estimators calculated for a stratum are always biased, while estimators for a domain are always unbiased.

MCQ4. In the context of ratio estimation, if x and y are positively correlated and the sample underestimates $\bar{x}_{\mathcal{U}}$ (i.e., $\bar{x}_{\mathcal{S}} < \bar{x}_{\mathcal{U}}$), how does the ratio estimator compare to $\bar{y}_{\mathcal{S}}$ and why?

(A) It will be higher, since the sample likely over-represents high-y units.

(B) It will be lower, since the sample likely over-represents high-y units.

(C) It will be the same; the ratio estimator only adjusts for t_y.

(D) It will be higher, since low x's likely implies low y's in the sample.

~~(E) It will be lower, since the sample likely over-represents high-y units.~~

MCQ5. A research is conducting a survey to evaluate household water usage across a large metropolitan area. The city is divided into 120 distinct residential zones (neighborhoods blocks). The researcher randomly selects 8 of these zones and selects every household within these zones to complete the survey. Across the 8 sampled zones, there are a total of 240 households surveyed. Which of the following correctly identifies the sampling design, the primary sampling units (PSU), the secondary sampling units (SSUs), and the value of m_i for each sampled zone i?

(A) Design: Two-stage cluster sampling; PSUs: Households; SSUs: Residential zones; $m_i = M_i$.

(B) Design: Two-stage cluster sampling; PSUs: Residential zones; SSUs: Households; $m_i = 30 = 240/8$.

(C) Design: One-stage cluster sampling; PSUs: Residential zones; SSUs: Households; $m_i = M_i$.

(D) Design: One-stage cluster sampling; PSUs: Residential zones; SSUs: Households; $m_i = 30 = 240/8$.

(E) Design: Two-stage cluster sampling; PSUs: Households; SSUs: Residential zones; $m_i = 30 = 240/8$.

MCQ6. Which statement correctly pairs the sampling design with 1) its group-level sampling rule and 2) **ideal** within-group composition (not what typically occurs)?

(A) **Stratified:** 1) Sample from SOME groups; 2) groups should be homogeneous inside.
Cluster: 1) Sample from ALL groups; 2) groups should be heterogeneous inside.

(B) **Stratified:** 1) Sample from ALL groups; 2) groups should be heterogeneous inside.
Cluster: 1) Sample from SOME groups; 2) groups should be homogeneous inside.

(C) **Stratified:** 1) Sample from ALL groups; 2) groups should be homogeneous inside.
Cluster: 1) Sample from SOME groups; 2) groups should be heterogeneous inside.

(D) **Stratified:** 1) Sample from SOME groups; 2) groups should be heterogeneous inside.
Cluster: 1) Sample from ALL groups; 2) groups should be homogeneous inside.

(E) **Stratified:** 1) Sample from ALL groups; 2) groups should be homogeneous inside.
Cluster: 1) Sample from ALL groups; 2) groups should be homogeneous inside.

MCQ7. In a one-stage cluster sampling design with equal cluster sizes ($M_i = M$), we select a simple random sample of n primary sampling units (PSUs) from a population of N total PSUs. Let y_{ij} define characteristic for the jth SSU in PUS i. The sample of PSUs is denoted as S and let I_i be defined as:

$$I_i = \begin{cases} 1 & \text{if PSU } i \text{ is selected in the sample} \\ 0 & \text{otherwise} \end{cases}$$

Which of the following statements is **false**.

(A) $\mathrm{E}[I_i] = \frac{n}{N}$

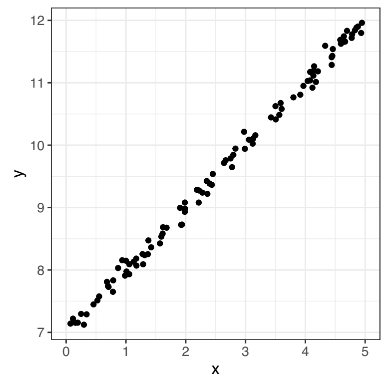
(B) The estimator for the population total is defined as $\hat{t} = \frac{N}{n} \sum_{i \in \mathcal{S}} \sum_{j=1}^{M} y_{ij}$

(C) The population total is defined as $t = \sum_{i=1}^{N} \sum_{j=1}^{M} y_{ij}$

(D) The estimator for the population total is defined as $\hat{t} = \sum_{i=1}^{N} \sum_{j=1}^{M} I_i y_{ij}$

(E) P(jth SSU in ith PSU is selected) $= \frac{n}{N}$

MCQ8. Based on the scatter plot of x and y, which estimator is more appropriate to estimate the population mean of y if t_x and N is known?



(A) Simple SRS mean $\bar{y}_{\mathcal{S}}$, because both the ratio and regression estimator rely on the population mean $\bar{y}_{\mathcal{U}}$ being known.

(B) Simple SRS mean $\bar{y}_{\mathcal{S}}$, because the correlation between x and y is weak.

(C) Regression estimator, because the proportionality assumption is violated.

(D) Ratio estimator, because the ratio estimator always has a smaller MSE than the regression estimator.

(E) Ratio estimator, because the relationship is linear.

MCQ9. Select the **false** statement. All else being equal, the variance of a ratio estimator for the population mean is smaller when

(A) The sample size n is larger.

(B) The observed (x, y) pairs lie closer to the ratio line $\hat{y} = \hat{B}x$.

(C) The relationship between x and y is stronger.

(D) The correlation between x and y is negative.

(E) The ratio of the population mean of the auxiliary variable, $\bar{x}_{\mathcal{U}}$ to the sample mean $\bar{x}_{\mathcal{S}}$, $\bar{x}_{\mathcal{U}}/\bar{x}_{\mathcal{S}}$, is smaller.

MCQ10. A simple random sample of $n = 100$ is taken to estimate mean income. After data collection, it is discovered that the sample contains too few young individuals relative to known population proportions. Suppose younger individuals tend to have lower average income than older individuals.

How will the post-stratified estimator $\bar{y}_{\text{post}}$ compare to the unweighted sample mean $\bar{y}_{\mathcal{S}}$?

(A) $\bar{y}_{\text{post}}$ will be smaller than $\bar{y}_{\mathcal{S}}$ because income observations of young individuals are up-weighted.

(B) $\bar{y}_{\text{post}}$ cannot be computed unless the original sampling design was stratified.

(C) $\bar{y}_{\text{post}}$ will be larger than $\bar{y}_{\mathcal{S}}$ because income observation of young individuals are down-weighted.

(D) $\bar{y}_{\text{post}}$ will be smaller than $\bar{y}_{\mathcal{S}}$ because post-stratification forces all post-strata means to be equal before averaging.

(E) $\bar{y}_{\text{post}} = \bar{y}_{\mathcal{S}}$ because SRS already guarantees representativeness in every subgroup.